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Modelling and simulation of MRP II activities in multi agent systems

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Modelling and simulation of MRP II activities in multi agent systems

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This paper is modelled in details, and it describes an integrated MRP II agent system for use in a make-to-order manufacturing environment by demonstrating potential benefits on purchasing and manufacturing orders generated. MRP II activities were modelled in a multi-agent based system; the information exchanges and activities to occur within the system were identified and the system simulation was prepared by applying the Petri net method using the estimated operation times for these activities. Multi-agent systems were preferred for modelling due to the fact that these systems were intelligent software systems and they included discrete manufacturing systems as well as communication and software systems. Also, the Petri net system was preferred in simulation because it was one of the distributed artificial intelligence methods and used in the analysis of the status and information exchange in the software systems. The obtained results will provide information about the possible bottlenecks and interruptions to occur before implemented within a huge and complex system structure.

Keywords: MRP II; Multi-agent system; Petri net; Simulation

1. Introduction

In this study, a material requirements planning (MRP) structure was modelled in MAS. Afterwards, this modelled system performance was evaluated in Petri net simulation, which was formed in MATLAB. As far as information and material flow was concerned, the criterion of time was used as a basis. Petri net is particularly preferred for the distributed system. For this reason, we preferred to apply the Petri net program for multi agent systems (MAS). The aim of modelling manufacturing resources planning (MRP II) activities in MAS structure was reviewed and the MRP system behaviours were analysed in detail. Such a simulation model was formed in order to ensure easy adaptation to rapidly-changing and competing market conditions. In particular, the fact that manufacturing, including assembly activities and the MRP system, was constructed in a huge structure, made minimising the lead time and cost in manufacturing inevitable to provide easy adaptation to rapid changing conditions. In this study, the multi-agent systems were preferred in modelling the system. Multi-agent systems use an object-oriented approach to artificial intelligence. Here, the aspects of the agent are defined in details. Multi-agent includes the interaction of the agents and data exchange activities. All the possible data exchanges to occur within the system were identified and described. Afterwards, the described system was simulated with the Petri net approach.

Section 2 includes the literature survey in MRP II, MAS and Petri net. Section 3 covers the MRP II model in MAS situations. Section 4 reviews the Petri net

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simulation approach and the last section evaluates the results and MRP II agent approach.

2. Literature review

In 1960, inventory control companies would manage to survive by satisfying the customer demand through controlling the inventory with a just-in-case attitude. However, in 1970, it became increasingly evident that companies could no longer afford the luxury of maintaining large quantities of inventory. This led to the introduction of MRP systems. MRP represented a huge step forward in the materials planning process. In 1980 companies began to take advantage of the increased power and affordability of available technology and were able to couple the movement of inventory with the coincident financial activity. MRP II systems evolved to incorporate the financial accounting system and the financial management system along with the manufacturing and materials management systems. By the early 1990s, continuing improvements in technology allowed MRP II to be expanded to incorporate all resource planning for the entire enterprise. Areas such as product design, information warehousing, materials planning, capacity planning, communication systems, human resources, finance, and project management could now be included in the plan (Oden *et al*. 1993, Ptak and Schragenheim 1999, Shankarnarayanan 2000). In this study, since MAS would provide agility to the system, this kind of method was preferred in modelling.

The MAS offers modularity. An open system is one in which the structure of the system itself is capable of changing dynamically. Such a system was characterised as a multi-agent system due to the fact that different software tools and techniques could be used by different people in different times. The MAS can be defined as a loosely coupled network of problem solvers that interact to solve problems beyond the individual capabilities or knowledge of each problem solver. These problem solvers, which are often called agents, are autonomous and can be heterogeneous in nature. The characteristics of MAS can be summarised as follows:

- Each agent has incomplete information or capabilities to solve the problem and, thus, has a limited viewpoint.
- There is no control over the entire system.
- Data are decentralised.
- The computation is asynchronous (Maturana *et al*. 1997, Shen *et al*. 1998, Sycara 1998, Aydın and Öztemel 2000, Ünver *et al*. 2000, Qiao and Zhu 2001, Shen *et al*. 2001).

In modelling dynamic structure of the agents, Petri net was used. The MAS were preferred as modelling because of the characteristics stated above. The Petri net approach was adopted to ensure evaluation of immediate status data and information exchange within the modelled system. Petri net is an artificial intelligence approach preferred especially in communication systems simulations. Petri net is commonly used in modelling and simulation of communication and software systems to represent the discrete event simulation modelling.

A time Petri net model was first proposed by Zuberek (1980). In this model, firing times, firing frequencies and transitions were described.

3. Suggested system

The objective of this study is to produce an output on-time with minimal cycle time and low work-in-process cost. The MRP II system allows entry of orders, process data, and design data, and performs most of the usual MRP II functions. The required output of the MRP II system is a dispatch report containing a queue of orders for each workstation with quantity of each part and other required information (Rembold *et al*. 1993, Sing, 1996). The CIM model consists of six functional sub-systems:

1. Order entry.
2. Computer-aided product design.
3. Process planning.
4. Production planning (MRP II).
5. Shop floor control.
6. Work in progress tracking.

Three major sub-systems of this CIM system, as shown in figure 1, are the process planning system, materials requirement planning (MRP II) system and shop floor control system (Pant *et al*. 1994). MRP II activities play an important role in the whole system.

The process planning system generates routing data and detailed data from shop reference data and design data. Routing data are concerned with a sequence of operations, and detailed data are related to information such as numerical control data, point position files, etc.

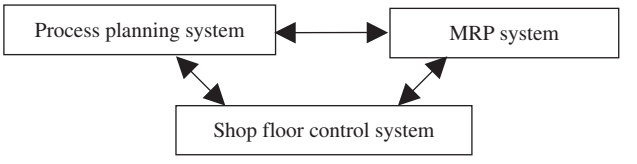


Figure 1. Major subsystems of CIM model (Pant *et al*. 1994).

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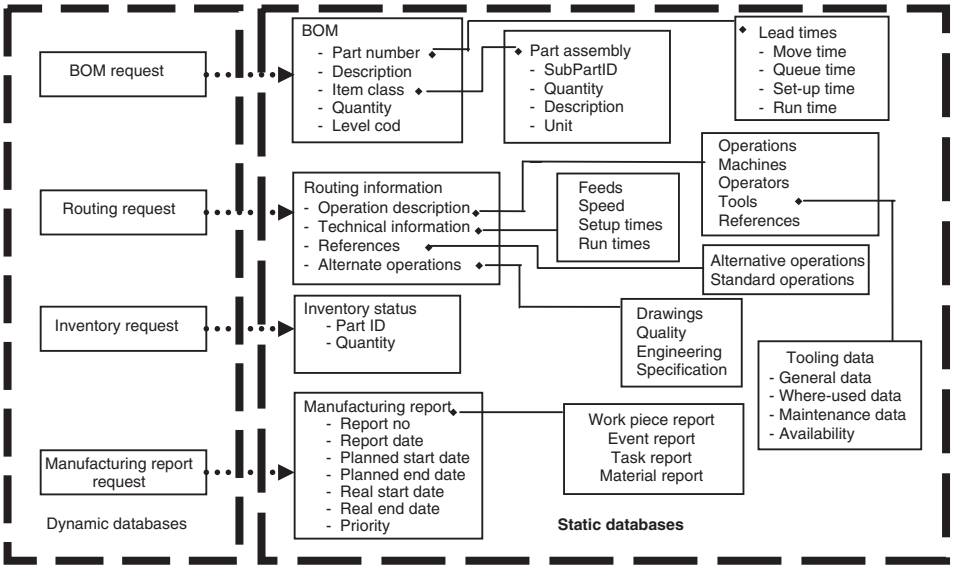


Figure 2. Designed MRP II databases and a part of relations.

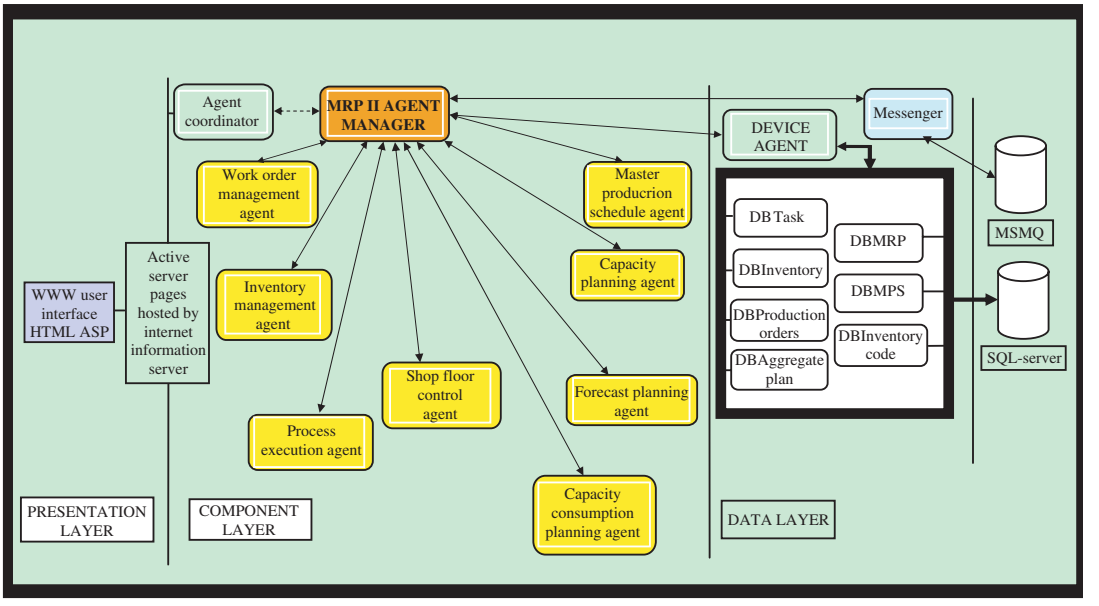


Figure 3. MRP II agent module.

The other principal output of the process planning system is confirmation of the availability of routing and header data in response to a query from the shop floor control system for process planning status. We discuss the MRP II sub-sections and database structure in figure 2. These sub-sections make up the modelled system components and events. This system would include all of the MRP II elements (Love and Barton 1996, Howard *et al.* 1998). We discussed the system elements in two sections. These sections are data and events consisting of dynamic and static structure. Dynamic structure refers to system events and static structure means database structure.

Once we prepared the system database, we regulated the relationship within the databases (figure 2). We designed the MRP II agent system shown in figure 3. This system contains all the MRP II activities (Kubat *et al.* 2002). This modular system includes computers.

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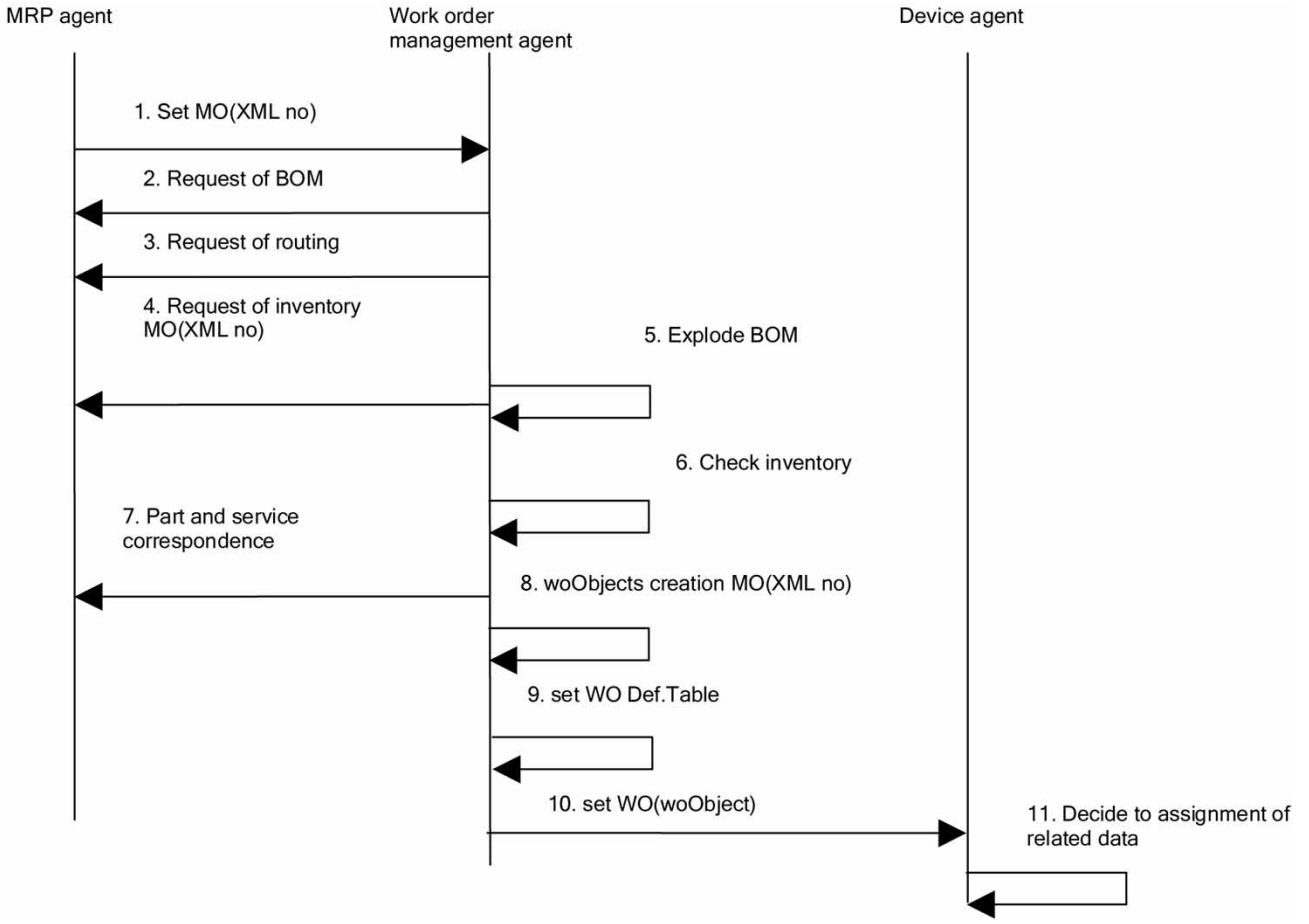


Figure 4. Detailed event diagram (Pab 4.2, 2002, Tsai *et al.* 2004).

The suggested system messaging procedure between two agents (the device agent and each of other agents) can be described as follows.

When the device agent receives a task waiting in the queue, it decides to execute a pre-task prior to the job, and sends a task announcement to each of the related MRP II agents. Other agents and MRP II agents are responsible for preparing bids, and they send the prepared bids to the message queue of the device agent. Any bid preparation schema as explained above can be built in each of the MRP II agents. After the device agent receives all the bids from its servers, it selects the agent with the lowest bid value (or preferably the highest, depending on the bidding schema) and sends it to the task-offer queue of the server. Each of the MRP II agents (related agent) reads the task offer and sends an 'accept' or 'reject' acknowledgement to the customer. If the response is 'accept', each of the other agents is selected and the bidding is finished. The device agent sends the task to the queue and waits for feedback at its task status queue. This flexibility allows the agents to communicate and select one to execute their own pre- and after-tasks. In this case, the bidding algorithm is a part of the manufacturing process to produce itself. The device agent would only send the task to that machine's task list and wait for the task status to complete its manufacturing. The built-in-agent systems and a number of selection procedures can be applied. We discuss the events and data. Events are shown in figure 4. The detailed event diagram shows the working attitude of the MRP II agent model. In table 1, each messaging and event processing is represented in Petri net.

4. The Petri net model

The use of Petri nets for describing and specifying the logic and controlling flow of computer systems has long proven itself invaluable for many fields related to

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Table 1. MRP II agent events list.

Transition no.	Description	From to/To from	Process time
1	SetMo(XML)	MRP II Agent to WO Manager Agent	20
2	BOM Request	WO Manager Agent to MRP II Agent	15
3	Routing Request	WO Manager Agent to MRP II Agent	10
4	Inventory Request	WO Manager Agent to MRP II Agent	12
5	Explode BOM	WO Manager Agent to WO Manager Agent	15
6	Check Inventory	WO Manager Agent to WO Manager Agent	15
7	Part and Service Correspondence	WO Manager Agent to MRP II Agent	15
8	WO Object Creation	WO Manager Agent to WO Manager Agent	15
9	Set WOIDinWIP Table	WO Manager Agent to WO Manager Agent	15
10	Set WO(WO Object List)	WO Manager Agent to Agent Manager	15
11	Migration	WO Manager Agent to Product Agent	8
12	Update WO Report	WO Manager Agent to WO Manager Agent	15
13	Terminate	WO Manager Agent to Product Agent	17
14	Terminate	Product Agent	6
15	Build MO Report	WO Manager Agent to WO Manager Agent	15
16	Transmit MO Report	WO Manager Agent to MRP II Agent	18
17	GetNOInfo (MO Information Request)	MRP II Agent to WO Manager Agent	40
18	GetWOList(moID)	WO Manager Agent to WO Manager Agent	15
19	Check WIP Table	WO Manager Agent to WO Manager Agent	15
20	GetWOInfo(woID)	WO Manager Agent to Agent Manager	85
21	FindPAsLocation	Agent Manager to Agent Manager	7
22	GetWOStatus	Agent Manager to Product Agent	70
23	Return the information	Product Agent to Agent Manager	90
24	WOStatusObject	Agent Manager to WO Manager	60
25	Input InWIP Table	WO Manager Agent to WO Manager Agent	15
26	CreatesMOInformationReport	WO Manager Agent to WO Manager Agent	15
27	MoInformationReport	WO Manager Agent to MRP Agent	50
28	GetInfo(XML infoRequest)	MRPAgent to Information Request Manager	45
29	Analysis	Information Request Manager to Information Request Manager	25
30	Create subject	Information Request Manager to Agent Manager	30
31	Creation	Agent Manager to Product Management Agent	20
32	Get URL	Product Management Agent to Look-up Service	20
33	Send Information	Look-up Service to Product Management Agent	18
34	GetData	Product Management Agent to Coord.Man.Unit	20
35	Information Report	Coord.Man.Unit to Product Management Agent	60
36	SetInfo(Information Report)	Product Management Agent to Agent Manager	90
37	Terminate	Agent Manager to Product Management Agent	15
38	Terminate	Product Management Agent to Product Man.Agent	18
39	Send Information	Device Agent to MRP II Agent	22
40	ConstructInformation Report	Information Request Manager to Information Request Manager	30
41	Reply	Information Request Manager to MRP Agent	35
42	GetPlantInfo (plantInformationRequest)	MRP II Agent to Device Agent	80
43	GetServiceList	Device Agent to Look-up Service	40
44	GetInfo( )	Device Agent to Coord.Man.Unit	20
45	Construct Report	Device Agent to Device Agent	12
46	plantInformationReport	Device Agent to MRP II Agent	10

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computer systems, such as communication protocols, processor pipelines and software algorithms (Agerwala 1979, Molloy 1989). One of the first models of Petri net with time was presented by Zuberek (1980) which associated deterministic firing times and firing frequencies to the transitions. However, the proposed solution to this model imposed some restrictions on the topology of the net. An extension to this model was later proposed by Holliday and Vernon (1987) and removed some of the restrictions on the topology of the net. Discrete event-distributed simulation usually assumes that the system to be simulated can spatially be partitioned into subsystems. Each subsystem is simulated by an associated logical process

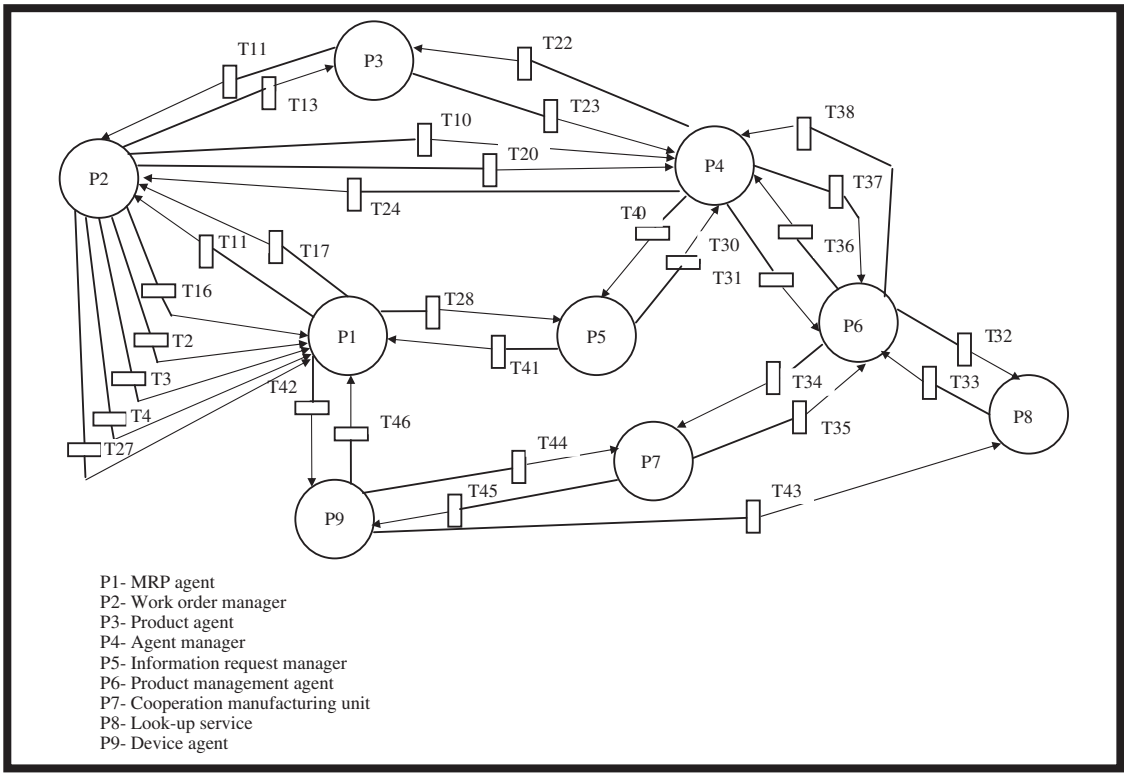


Figure 5. Petri net presentation of the activities in MRP II agent model given in table 1.

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(Nketsa and Valette 2001). This conservative distributed simulator of Petri nets is presented in Nketsa (1994a, b) and Khalifa *et al*. (1995). It defines a framework to build distributed simulation of any types of Petri nets. The simulated Petri net has to be decomposed in modules or in levels which are assigned to a computer of the network.

We are going to study how the assignment of subnets to different tasks (or local processors) can affect the implementation via simulation of the system. We are trying to evaluate the agents introduced by the distributed of the system structure.

The approach adopted in this article is based on three steps:

1. Petri net modelling in convenient graphical design.
2. Export to matrix representation in MATLAB.
3. MATLAB-based analysis program used and the obtained simulation result shown later in figure 6.

When a connected system is made by system specification, the interim Petri net model of the connected system is generated by MATLAB program, which is based on a timed coloured Petri net.

Definition of Petri net (P, T, A, x_0):

$$P = [p_1, p_2, \ldots, p_n] \tag{1}$$

where P is the set of places (representing conditions or number of parts). T is the set of transitions (corresponds to events),

$$T = [t_1, t_2, \ldots, t_m]; \tag{2}$$

$$A \subseteq (P \times T) \cup (T \times P) \tag{3}$$

is a set of arcs from places to transitions and from transitions to places, and

$$x = [x(p_1), x(p_2), \ldots, x(p_{n1})] \in N^n \tag{4}$$

is connecting p_i to t_j. When a transition fires, the vector of markings changes:

Definition of firing in ordinary Petri net
A transition $t_j \in T$ fires in an ordinary Petri net, if and only if $x(p_i) \geq w(p_i, t_j)$ for all $p_i \in I$ (t_j), changes occur in markings

$$x'(p_i) = x(p_i) - w(p_i, t_j) + w(t_j, p_i), \quad i = 1, \ldots, n. \tag{5}$$

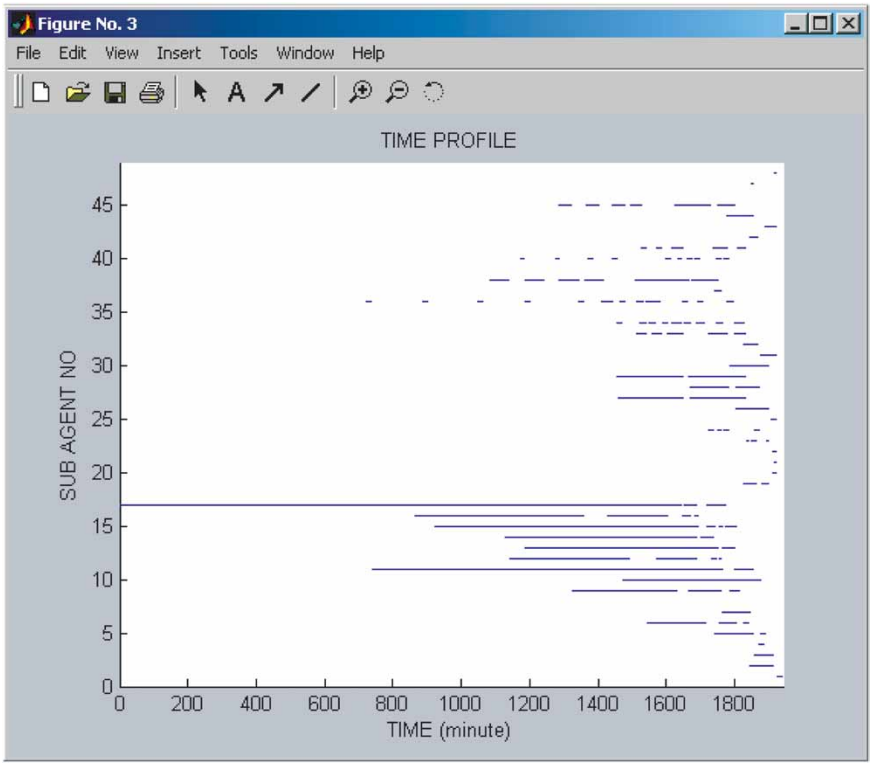


Figure 6. Results of Petri net simulation in Matlab and all possible operation periods were defined.

Simulation was made with the use of Petri net model of the MRP II agent activities given in figure 5. In simulation, the calculation was performed by representing the activities in the incidence matrix. The rules for generating Petri net model from the connected system are given in figure 5. Petri net model in a matrix form is called the incidence matrix. The incidence matrix representing a Petri net is actually a matrix of input places and output places of the transitions in the Petri net model (Tricas and Martinez 1998, Park *et al*. 2001, Der *et al*. 2004). The Petri net model works by coupling with the systems elements. Defined status and coupling points of the system and simulation were realised in the MATLAB environment under the following rules:

1. Each agent is replaced by transition-place pair. The transition represents the (overall) production at the specific agent and the place represents the out-buffer for products of that enterprise.
2. Each transportation is replaced by a transition-place pair. The transition is connected to the agent ('from' agent), and the place is connected to a different agent ('to' agent). The weight of the arc between the transition and the place is the batch size.
3. The weight of the arcs between the places of the transition and the agent transition is the component ratio. Component ratio means how many units of parts flowing through transportation are needed to make a unit of product and information at the related agent (Davidrajuh 2000).

4.1 *Simulation with Petri net model*

We chose the Petri net system model for simulations because we wanted to make sure that the simulations (time calculation) would be done in linear time. Petri net ensures that the methodology used to operate the Petri net model in calculations will be done in linear time, and that $T(j)$ is the number of transportation. This requirement on linear time is a necessity. Otherwise, because of the size of the system model (large number of agents), non-linear simulation time will make the implementation inappropriate as an on-line real-time monitor. Calculation is done in one traversal. The operation manner of the program includes information both on

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Table 2. Program algorithm (adapted from Zuberek 1980).

```
[Seq,M] = incidence matrix(P, T, M0, TimeT, mark)
% P- Places, T- Transitions, M0- Vector of initial markings,
% mark- Number of simulation mark, Seq- Firing Sequence,
% M- Marking at the end of the simulation
M = M0;

For j = 1: number of transitions
  CountT(j) = initial counter value
End

While t < = ticks
For j: = 1: number of transitions
  If transition j is enabled
        Then x(j) = 1
              If CountT(j) = 0   then y(j) = 1
        Else y(j) = 0
              End
        Else x(j) = 0
              CountT(j) = initial counter value
  End
Else
  , f = randomly chosen fireable transition in y
  M = M + T(:f) − P(:,f)
  % add fired transition to firing sequence
  Seq = [Seq,f]
End
End.
```

data flow and material flow operations. This reversing is done simply by swapping the input places and output places in the incidence matrix. The program algorithm is given in table 2.

All messaging among MRP II agents and their estimated processing times is provided in table 1. The study will provide information about the working mechanisms of the system under these estimated processing times. This will be done by use of a simulation. The results of Petri net simulation in MATLAB are given in figure 6. The figure shows the Petri net simulation results as a time schedule which will be used to make decisions.

5. Conclusion

It was observed that the information exchange between the MRP II Agent and WO Manager Agent became important. The system simulation was realised after all possible MRP II activities within the system modelled with MRP Agent were identified. MRP II has its strength in job shops requiring flexibility in the production sequence, in the quality of production and in timing of the production process.

Since the MRP II concept is of great importance for the competitive conditions of companies, the system was modelled and simulated in an agent environment. As a result of this study, system optimal decisions, (i) where to produce, (ii) how to distribute the product and (iii) at what time to order or deliver, can successfully be discussed and evaluated in a transformed environment. The capacity selection and process selection, for instance, can more easily be discussed in a transformed environment.

As a result of the simulation based on the information given earlier in table 1, it is observed that the activities within the event 10–25 range have an essential place within the system in terms of time. It is also observed that the information exchange between MRP Agent and WO Manager Agent, the activity numbered 17, is especially important. In the model of the system in this study, events 10 and 25 transitions are important. When the system is realised, figure 6 especially will lead to system status in the future. The 17th activity (GetNOInfo(MO Information Request) MRP Agent to WO Manager Agent) plays a main role in the system. This activity has a greater importance because it used a larger time span than other activities within the system.

Table 1 provides information on how the system works. For example, it indicates that there is an extensive information flow between events 10 and 25. This means there are going to be bottlenecks and queues in the units responsible for events 10 and 25.

The process and capacity relation is also taken into consideration in this study. Control is based on aggregate measures of summed processing times. The central release function balances loads across capacity groups. The system simulation framework proposed by us contributes to:

Arrival	Arrival density, arrival time and variability in system.
Due dates	Due date density and due date allowance variability.
Operations	Processing time groups, processing time variability and setup/processing time ratio.
Routings	Routing sequence variability, routing length, routing length variability, routing flexibility and level of convergence.

In conclusion, this study allows us to get an idea about the types of events and times defined in the system in this study with the help of such a simulation model. This model will help us to learn about the possible bottlenecks and interruptions to occur within huge and complex systems via a preliminary work before the implementation of the system. Finally, the study indicates how a multi agent system simulation can help decision making about the processes.

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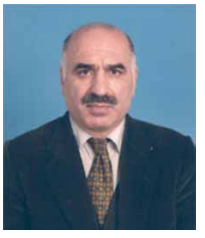
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